Energy-Efficient Algorithm to Construct the Information Potential Field in WSNs

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***Abstract*— The information gradient-based routing and navigation protocols have been proved to be effective when collecting data from the distributed wireless sensor networks, because the data collector can achieve the global objective through local greedy decision based on the information gradient. An efficient method to establish this information gradient is to solve a discrete approximation to the harmonic function, which is called information potential field (IPF). However, the energy-efficient and quick convergence methods to construct the IPF should be fully investigated to trade off the energy efficiency and the quality of the IPF, especially in the large-scale networks with high dynamics. In this paper, two algorithms are proposed to efficiently construct the IPF, including *hierarchical skeleton-based construction algorithm* and *value estimating substitution algorithm*. Both algorithms obey the typical hypotheses on WSN settings and the gossip-styled propagation principle. In addition, we propose the advanced approaches to construct the IPF to tackle the challenges of its practical applications, such as obstacles, task priorities, and sensor energy budget. Comprehensive simulation results show the feasibility of the proposed algorithms, which can reduce the number of iterations to reach a convergence status by 80% so as to conserve energy, and they perform well considering the requirements of real applications.**

***Index Terms*— Information gradient, information potential field, iteration convergence, energy efficiency, greedy routing.**

I. INTRODUCTION

FOR the traditional data collection protocols using static sinks in wireless sensor networks (WSNs), the energy of sensor nodes near the sink will be depleted dramatically that shortens the lifetime of the network. Recently mobile sinks have been introduced to collect data and to effectively achieve the goal of extending the lifetime of the network through balancing the energy consumption of the entire network [1]. An example application using mobile sinks is the forest fire detection system, where numerous sensors are deployed to detect fire emergency based on the temperature and smoke concentration, which are related to the fire emergency event. Mobile sinks with fire-fighters move around the forests to collect the data from the sensor nodes and data-driven decisions effectively navigate fire-fighters to the fire emergency spots.

However, the mobility of sinks can result in unexpected changes in the destinations of routing, which can largely affect the delivery performance of emergency events to sinks [2], [3], and consume a lot of extra energy to update the routes. Therefore, the information gradient-based routing methods are proposed to navigate the mobile sinks [4]–[8]. In their designs, data of events of interest (EoI) are diffused across a virtual information gradient, and sinks move along the gradient of a specific EoI to the location of EoI. There are two types of the information gradient, the natural information gradient and the artificial information gradient. The former imitates the spread process of physical parameters such as heat and light. It may have local extremums in the resulting gradient [4] which can lead to a random walk in navigating mobile sinks [5]. Many efforts also have been made to build artificial information gradients. Researchers adopt the hop count to build the information gradients [6], but they need frequent flooding to guarantee the accuracy of the gradient when network topology changes.

Lin *et al.* proposed to create an artificial information gradient that is robust and has no local extremum [7]. In their definition, the artificial information gradient is called the *Information Potential Field (IPF)*. They use the harmonic function to build IPF, which represents the diffusion of a particular type of EoI such as fire emergency from the source of EoI. The IPF that simulates a harmonic function also has multiple advantages due to its nice algebraic properties, but the cost of establishing such a IPF is prohibitively high.

This work aims to solve the problem of developing energy-efficient algorithms to construct IPF while considering the practical issues of applying IPF in real applications. We first examine the trade-off between the convergence speed and the energy consumption, especially in a large-scale network scenario where EoIs appear and disappear dynamically. Next we propose two algorithms to construct the IPF: 1) Hierarchical Skeleton-based Construction Algorithm (HSCA) and 2) Value Estimating Substitution Algorithm (VESA). Both algorithms are validated to be energy-efficient and perform well in large-scale networks through our extensive

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TABLE I
NOTATIONS

Symbol	Definitions
$v(i,j)$	A sensor node on the position (i,j)
$N(u)$	The set of neighbour nodes of node u
$d(u)$	The connection degree of node u
$\Phi(i,j)$	The information intensity value (IIV) of node in $v(i,j)$
$\Phi'(u)$	The new IIV of node u computed by the Jacobi iteration based on the current value
$\Phi(u)$	The IIV of node u
ε	The update threshold for updating the IIV
δ	The pre-specified threshold for average proportion of the IPF difference of the ideal information potential
E_{cmp}	The energy consumption for conducting an iteration computation
E_{itr}	The energy consumption for message interaction between two nodes
$T_u(\varepsilon, \delta, size)$	The number of iterations of node u, which is a function of the three variables including ε, δ and network size.

experiments. Furthermore, there are several practical issues when IPF is applied in real applications, such as multiple EoIs, prioritized EoIs, obstacles in the sensing area, and low energy sensors. To deal with above problems, we extend the definition of the IPF with priority to meet different levels of delay-tolerant properties of EoIs and composite IPF to navigate sinks to avoid obstacles and bypass low power nodes.

In summary, the contribution of the paper is three-fold.

- We demonstrate the trade-off between energy efficiency and the quality of the IPF in the construction of IPF, especially in large-scale WSNs.
- We propose two energy-efficient algorithms for constructing an ideal IPF by accelerating the convergence speed of iteration.
- We extend the definition of IPF by considering the needs of real life applications and conduct comprehensive performance evaluation to validate the efficiency and effectiveness of our proposed algorithms.

The rest of this paper is organized as follows. In Section II we introduce the motivation and present formal models concerning the problem. In Section III we propose two energy-efficient algorithms for constructing the IPF, which is followed by the extended definition of IPF. An example of the IPF application illustrated in Section IV. Comprehensive simulation results are presented in Section V and the list of related work is indicated in Section VI. Finally, the paper is concluded in Section VII.

II. Preliminaries

This section first introduces the motivation of this paper, and then presents the network model. Moreover, the problem is formulated. All notations are defined in Table I.

A. Motivation

The work [7] provides an information gradient scheme, which is the *Information Potential Field (IPF)* established by

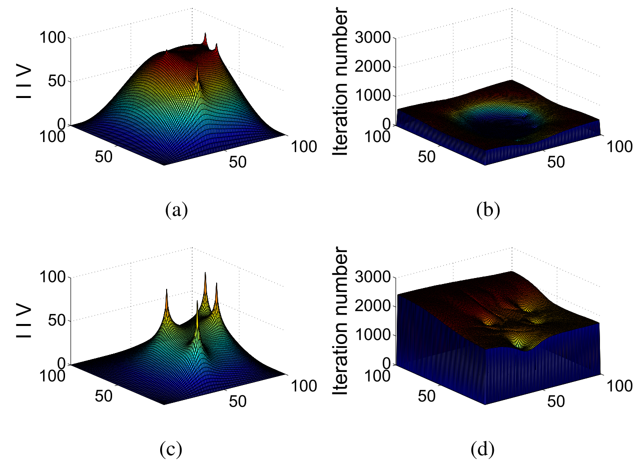


Fig. 1. The tradeoff between the quality of IPF and the cost (each node's iteration number) in a large-scale network by using the Jacobi iteration method. (a) IPF when $\varepsilon = 0.01\%$. (b) Cost when $\varepsilon = 0.01\%$. (c) IPF when $\varepsilon = 0.001\%$. (d) Cost when $\varepsilon = 0.001\%$.

the Jacobi iteration. Each node holds an *Information Intensity Value (IIV)* that indicates the diffused strength of EoI, which is diffused away from source nodes. The non-source node receives the IIV from its neighbors and performs the Jacobi iteration to compute the IIV of this node. The value update of one node will trigger its neighbor nodes to iterate. To reduce the number of iterations, the update constraints are proposed in [7]. The IIV of node u is updated only if the new value $\Phi'(u)$ is sufficiently different from the old value $\Phi(u)$, i.e., the update stops if $|\Phi'(u) - \Phi(u)| \le \varepsilon\{\Phi'(u), \Phi(u)\}$. In this way, the values of the IPF is approximate to a harmonic function, but the method described above may cause the non-convergence problem in a large-scale network. The smaller ε we choose, the better the resulting values of the IPF approximate to a harmonic function but the more energy will be consumed. This is because that an iteration operation costs a large amount of energy to request information from its neighbors and perform computation. In other words, how to balance the degree of approximation to the harmonic function and the energy consumption needs to be explored extensively.

The energy overhead for iteration and non-convergence problem are not obvious in a small network. However, when the number of sensors increases, we can witness the trade-off between the quality of IPF and number of iterations in a large-scale network by using the Jacobi iteration method, which is described in paper [9]. We validate this argument using a simple experiment in a 100×100 grid network with four sources. The IIV of sources is set as maximum of 100, and the IIV of perimeter nodes is set as minimum of 0. As shown in Fig. 1a and Fig. 1b, although the energy consumption is low when we set the relative difference threshold to be $\varepsilon = 0.01\%$, the flat regions on the surface of the IPF appear. The flat region means that the IIVs of many nodes are very close. The local extremum refers to the non-source node holding a higher IIV than all neighbors also exists. The flat regions and the local extremum result from the chosen value of the epsilon. Both the local extremum and the flat region lead to random walking in navigating the movement of mobile sinks [7]. Indeed, further experiments show that when we choose a tighter threshold, such as $\varepsilon = 0.001\%$, the IPF approximates the harmonic function much better, as shown in Fig. 1c. However, it needs much more iterations so that it consumes prohibitively higher energy, as shown in Fig. 1d. We define this situation as a non-convergence problem. In summary, we may not have a high quality IPF if ϵ is loosely set, namely non-convergence problem. On the contrary, a tight ε will cause a significant rise in the number of iterations as well as in the data transmission, resulting in an increased energy consumption.

A comprehensive and effective protocol for navigating the mobile sinks should be designed to solve many fundamental problems (e.g., ad-hoc topology, energy efficiency) associated with practical applications. The ideal IPF closely approximates a harmonic function has no local maxima, and thus the IPF can be used to navigate mobile sinks move to EoI as quickly as possible by simply climbing the steep edge on the surface of the IPF, i.e., for each step of the mobile sink. The mobile sink moves to a neighboring node that has the maximum difference in the IIV. The IPF supports a greedy routing algorithm by simply ascending the smooth surface of the IPF. However, considering the practical issues that can block the movement of the mobile sink, such as obstacles and inaccessible areas, the concept of IPF could be extended. In addition, prioritization of EoIs and the avoidance of low power nodes need to be considered in the construction of the IPF.

B. Network Model

We denote the network as a 2-dimensional rectangle area of length l and width h having nodes inside and on its perimeter. The network contains a large number of sensor nodes, each node is powered by an unreplaceable battery. Without loss of generality, for convenience of simulation and analysis, we set the sensor network as a grid network, where the distance between each pair of adjacent sensors is d. Then, we model the network as an undirected graph $G = (V, E)$, each node $v(i, j)$, where (i, j) indicates the location of the node in the graph, represents a sensor. The sensor readings are categorized into a set of high-level events, e.g., the high temperature and high concentrations of smoke indicating the fire emergency event. The nodes which sense the EoIs are called *sources*. The EoIs may emerge anywhere at any time in the network and will disappear once they are reached by queries, so that sources experience a dynamic emergence and disappearance process.

We intend to establish the IPF that approximates the harmonic function. In a domain $D \subset R^2$, the *harmonic function* is a real function Φ which satisfies the Laplace equation [10]. The Jacobi iteration method is used to compute the discrete harmonic function in [7] to build the IPF. The IIV at each node forms the discrete IPF. According to the Dirichlet boundary conditions [11], the IIVs of the perimeter nodes are pre-specified to be of strength 0. The sources are also set as maximum IIV of 100. The IIVs of other non-boundary nodes are obtained by the Jacobi iteration, as shown in Equation (1). Each sensor only needs to communicate with its neighbors so that most data transmissions are avoided.

$$\Phi^{k+1}(u) \leftarrow \frac{1}{d(u)} \sum\nolimits_{v \in N(u)} \Phi^k(v) \tag{1}$$

C. Problem Formulation

In this paper, we aim to achieve the following design goals: reducing the number of iterations as much as possible when constructing the IPF, constructing the ideal IPF to approximate harmonic functions, and shortening the time to navigate the mobile nodes to the event source, while reducing energy consumption to prolong network lifetime.

1) Energy Consumption Model: Assuming that the energy consumption of each node is $E = E_{cmp} + E_{itr} \approx E_{itr}$, where the energy cost of iteration computation E_{cmp} is far less than that of message interaction E_{itr}. Hence, we mainly study how to reduce the energy consumption for message transmission in this paper. Each node computes the new IIV once it receives the IIV from any neighbors and dispatches the updated new value to all of its neighbors if the new IIV is sufficiently different from the old one. Adopting the message format defined in TinyOS [12], the message is about 64 bits including the node ID, the counter value and data. According to [13], the energy dissipated for the transmitter (or receiver) is $64bit \times 50nJ/bit = 3.2\mu J$, and the energy cost for the transmit amplifier is $64bit \times 100pJ/bit = 6.4nJ$. Ignoring the packet loss and feedback, the total energy consumption for the node is shown in Equation(2).

$$E \approx T_u(\varepsilon, \delta, size) \times (3.2\mu J + \sum\nolimits_{v \in N(u)} 6.4nJ) \tag{2}$$

It is required that the constructed IPF should approximate harmonic functions well, i.e., the degree of the approximation of the IPF to the harmonic function should meet an acceptable threshold. To give the quantified measurement, we give the following definitions of some parameters.

Definition 1: Ideal IPF: the IPF will be ideal if it satisfies the following conditions: 1) no local extremum or plateau regions; 2) the APIPD is less than a pre-specified threshold δ.

As mentioned in paper [7], the plateau regions are the place where the neighbors have the same IIV. This may result from either the features of harmonic function or network topology, and bring the random walk or flooding to the MDCs in further greedy routing, and the discussion of this issue also addressed in [14].

Definition 2: Proportion of Information Potential Difference (PIPD): the IIV $\Phi'(i, j)$ is computed by our algorithms, and the accurate value $\Phi(i, j)$ is obtained by solving the Laplace Equation. The PIPD in node $v(i, j)$ is defined in Equation(3).

$$PIPD_{(i,j)} = \frac{|\Phi'(i, j) - \Phi(i, j)|}{\Phi(i, j)} \tag{3}$$

Definition 3: Average proportion of Information Potential Difference (APIPD): the *Average Proportion of Information Potential Difference* for the whole network is given by

Equation(4).

$$APIPD = \frac{\sum_{i=1}^{h}\sum_{j=1}^{l} PIPD_{(i,j)}}{l \times h} \tag{4}$$

III. Energy-Efficient IPF Construction

As mentioned above, we intend to balance energy conservation and iteration convergence by optimizing two important performance indicators, i.e., the number of iteration and the number of messages. Therefore, we design two energy-efficient algorithms, Hierarchical Skeleton-based Construction Algorithm (HSCA) and Value Estimating Substitution Algorithm (VESA), to accelerate convergence of iteration.

A. HSCA

To better understand the HSCA, we first define some key terms.

Definition 4: **Skeleton nodes** are a part of nodes separated from each other by a certain hop-count in the network.

Definition 5: **Ordinary nodes** are all other nodes except the skeleton nodes in the network.

Definition 6: **Skeleton network** is composed of the whole skeleton nodes, at which the Jacobi iteration is performed.

Definition 7: **Skeleton granularity** is the hop that counts the distance between a pair of neighboring skeleton nodes.

The traditional Jacobi iteration method has to trade off energy consumption and iteration convergence when constructing the IPF, especially in the large-scale network where it is hard to balance the quality of the IPF and the energy cost for iterations.

Algorithm 1: HSCA

Input: The IIV of sources and perimeter nodes, and the location of sources.

Output: A qualified IPF of the whole network.

1: Selecting the proper skeleton nodes.

2: Iteration on the skeleton network: we perform the Jacobi iterations on the skeleton network until convergence appears. After this step each skeleton node holds a convergent IIV.

3: Interpolation: we apply the interpolation method to extend the IIVs from the skeleton network to the whole network.

4: Smoothing process: we run the Jacobi iteration in the whole network until we get the final qualified IPF .

The HSCA is shown in Algorithm 1. In step 1, we determine the skeleton granularity and select the skeleton nodes. Relay nodes are necessary for long-distance transmission [15], but the data transmission frequency of relay nodes is far higher than that of the other general nodes. The energy consumption makes the relay nodes become the bottleneck of the network lifetime.

Fig. 2 shows a simple example of the HSCA, where arrows specify the information flow at each step. We firstly perform the Jacobi iteration, which consists of nodes $S_{ci}(i \in [1, 4])$ to achieve the convergent IIVs at the nodes. Values at nodes in the

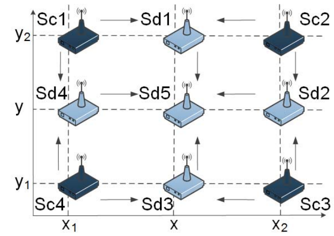


Fig. 2. Information flow schematic diagram. The arrows show the information flow between sensor nodes.

skeleton network map to the whole network unchanged, and the values of the skeleton nodes are kept unchanged. In detail, the IIVs at nodes $S_{di}(i \in [1, 4])$ are obtained by applying the Lagrange interpolation method [16] and the value of S_{d5} is computed through the bilinear interpolation method [17]. The last step is to perform iterations in the whole network, including the skeleton and ordinary nodes, until the ideal information field is obtained.

In the interpolation phase, we have two cases to handle, the 1D case and the 2D case. In the 1D case, the ordinary nodes and their skeleton neighbors are placed in a line. With the known IIV of its neighbors, the Lagrange interpolation method is adopted to calculate the information density value of ordinary nodes. For example, in Fig. 2, assumed that the IIV of skeleton nodes S_{c3} and S_{c4} are known, then the IIV of node S_{d3} can be calculated by Equation(5).

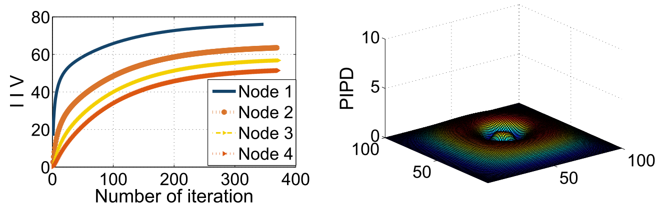
$$\Phi_{(S_{d3})} = \frac{\Phi(S_{c4})(x - x_2)}{x_1 - x_2} + \frac{\Phi(S_{c3})(x - x_1)}{x_1 - x_2} \tag{5}$$

In the 2D case, the ordinary nodes and the skeleton nodes are placed on a plane. For example, in Fig. 2, S_{d5} is the ordinary node, while S_{c1}, S_{c2}, S_{c3} and S_{c4} are the neighbor skeleton nodes of S_{c5}. With the values of these four skeleton nodes are known, the value of S_{c5} can be calculated with bilinear interpolation as shown in Equation(6).

$$\begin{aligned}\Phi_{(S_{d5})} \approx\ & \frac{\Phi(S_{c1})(x_2 - x)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)} + \frac{\Phi(S_{c2})(x - x_1)(y - y_2)}{(x_2 - x_1)(y_2 - y_1)} \\ & + \frac{\Phi(S_{c4})(x_2 - x)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)} + \frac{\Phi(S_{c3})(x - x_1)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)}\end{aligned} \tag{6}$$

Theorem 1: The HSCA can accelerate convergence of iteration during the IPF construction phase.

Proof: In our scheme, the harmonic function is used to simulate the diffusion of the EoI, so that the value of the IPF is obtained [7]. The harmonic function is a second derivatives function and meets the Laplace function. The Jacobi iteration is performed to obtain the discrete solution of the Laplace function. From the viewpoint of mathematics, the convergence speed and accuracy of iteration are limited by two modes of error, high-frequency oscillatory modes and low-frequency smooth modes [18]. According to [19] by performing iteration on a single grid only oscillatory modes of error can be eliminated effectively, but smooth modes are damped very slowly. This seems like a limitation, but we can use the smoothing property to advantage by using the sparse grids. Sparse grids can be used to compute an improved initial iterative value



(a) (b)

Fig. 3. (a) The iteration trends of four randomly selected nodes. (b) The PIPD by using VESA.

guess for fine-grid relaxation. This is an advantage because of the following reasons: 1)Relaxation is much cheaper and the number of nodes that need to calculate iteration values is much less in the sparse grid; 2)Smooth error is relatively more oscillatory on the more sparse grid [19]. For example, the error with frequency $k = 4$ is treated as smooth modes on fine-grid $\Omega^h(N = 12)$ for $k < (N/2)$, but appears to be high-frequency on the sparse grid, that is $\Omega^{2h}(N = 6)$ for $k > (N/2)$. Thus the multi-grid method can accelerate convergence of iteration. That is to say, the more sparse grids accelerate convergence of a relaxation scheme on finer grids by efficiently liquidating smooth error components, and fine-grids improve accuracy by correcting their forcing terms.

The HSCA is an application of the multi-grid method. We firstly perform Jacobi iteration on the skeleton network (more sparse grid), and skeleton nodes' convergent IIV can be used to give an improved iterative initial value guess through the interpolation method. Finally, a small quantity of iterations are executed in the entire network (finer gird) before iterative convergence. Thus the HSCA can accelerate convergence of iteration during the IPF construction phase.

B. VESA

The VESA can accelerate the convergence of iteration, and it is especially suitable for the case where the sources are dynamically and randomly deployed. For instance, the old source nodes may suddenly abort their information or new events of interest emerge.

Fig. 3a shows the trace of the IIV recorded by four sensors as the Jacobi iteration is executed. We sample the IIV of Node 1, Node 2, Node 3, and Node 4, with the increased distance to the source. Note that each curve in Fig. 3a represents a single node's IIV trace in correspondence to the number of iterations. For each node, we can learn that the IIV of each node increases as the iteration goes on, and that all four nodes exhibit a pattern of long tail iteration, which means the change of the intensity value after a set number of iterations is not distinct. Thus the convergence value of information density at each sensor node can be estimated by learning its historical IIV that is stored locally. The detailed implementation of the VESA is shown in Algorithm 2.

The main idea of the VESA can be explained as follows. We use the Jacobi iteration as a base operation. However, instead of the ceaseless computation on averaging the neighbors' IIVs, we predict the final converged value by analyzing the former iteration data. Currently, we use various prediction

Algorithm 2: VESA

Input: The IIV of sources and perimeter nodes and the location of sources.

Output: A qualified IPF of the whole network.

1: Perform the Jacobi iterations for m times in the network. Each sensor node records a series of data including m historical IIVs resulting from m iterations.
2: Based on the data sets from step 1, we calculate the predicted value for each node by the cubic extrapolation method.
3: Each node uses the predicted value to substitute the current IIV.
4: If no nodes need to be updated, exit; else go to step 1.

methods, including the linear extrapolation [20] and cubic extrapolation method [21] to predict the convergence value. Considering the computing cost and the accuracy demand, we infer that different extrapolation methods can fit with different scenarios.

In the 30×30 grid network, we select a tolerable threshold for APIPD as δ=0.005. An ideal IPF constructed by the VESA with APIPD equals to 0.0372. Fig. 3b shows the PIPD at each node. We notice that all of the PIPDs are very small, which shows the good performance of the VESA.

IV. Advanced Approaches of Constructing IPF

We have introduced how to construct the IPF in an energy-efficient manner. The IPF could navigate the mobile sinks to the sources. Mobile sinks can make decisions greedily based on the constructed IPF. In this section, we discuss how to construct the complex IPF by considering some real application scenarios, i.e., IPF., the prioritized IPF and composite IPF. The routing path also considers the residual energy of nodes which contributes to balance the load of each node.

A. Prioritized IPF

In case of multiple EoIs, the mobile sink reaches them in turn. However, multiple-events may have different levels of delay-tolerant properties because of their timeliness requirements, and different levels of urgency. We assume that all the events reported by the source nodes have the same event types. For example, in the fire monitoring application, there are multiple fire events happening in the same time. The fire keep active before they are handled. Because the economic loss caused by the emergencies are different, these events will be assigned with different levels and higher level events should be handled earlier. We assume that the level of the event can be quantitative.

Here, we adopt the logistic function to model the influence factor $\Theta(level)$ of different priority levels, shown in Equation(7).

$$\Theta(level) = \frac{1}{1 + e^{-level}} \tag{7}$$

where $\Theta(level)$ is an incremental value within the interval [0, 1].

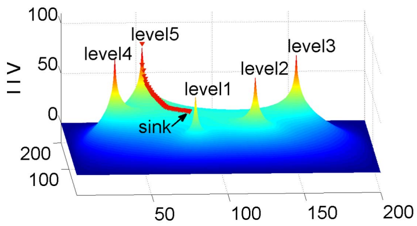


Fig. 4. The IPF with five different priority levels, which are classified into five priority levels (1,2,3,4,5).

The routes of the mobile sink are determined by the force towards the source generated from the IPF. The mobile sink at node u is required to move to one of its neighbors. The mobile sink just moves to the neighbor with the most significant difference in information density value. Considering the priority of different sources, we set the initial intensity value of multi-sources according to their influence factor $\Theta(level)$ of priorities. Assume that the maximum value of the IIV is denoted by max, the IIV of a certain source can be quantified by Equation(8).

$$value = max \times \theta(level) \tag{8}$$

We give an example to illustrate how to construct the IPF with priority and how it navigate mobile sinks. In a 200×200 network, there are five source nodes with different levels of priorities randomly deployed. By setting max to be 100, the IIVs of these five sources are 73.1059, 88.0797, 95.2574, 98.2014, and 99.3307 respectively. The IIV of the border nodes is fixed at 0. Both Algorithms 1 and 2 can be adopted to construct the prioritized IPF, as shown in Fig. 4. It is observed that the prioritized IPF provides mobile sink with the stronger guidance attracted to sources which carry the higher level of priorities. The initial position of the mobile sink, $(82, 79)$, is in the middle position between source $(24, 80)$ and source $(80, 30)$. The red triangles in Fig. 4 show the mobile sink's routing path, and it is observed that the mobile sink routes to sources which have a higher priority level.

B. Composite IPF Construction

In fire monitoring applications, there are also obstacles in the monitoring area, while mobile sinks cannot move through the obstacles but bypass them to reach the source. Here we define the inaccessible region as an *obstacle*, such as big stones, walls, trees, pits and other unreachable areas. To tackle this problem, we propose the composite IPF, in which the information of source events is integrated with that of the obstacles. In other words, each sensor $v(i, j)$ holds three data, the first one being the IIV of sources $\Phi_{(source)}(i, j)$, the second one being that of obstacles $\Phi_{(obstacle)}(i, j)$, and the third one being the composite IIV $\Phi'(i, j) = \Phi_{(source)}(i, j) + \Phi_{(obstacle)}(i, j)$. The composite IIVs of all nodes in the network form the composite IPF.

The composite IPF can make the route of the mobile sink to the source automatically avoiding obstacles because: 1) the opposite IPF diffused from the obstacles will reject the mobile sink down to the position of obstacles; 2) the positive IPF of the sources still attract query up to source vicinity.

The data diffusion scheme of the obstacle is similar to that of sources. The construction steps of the positive IPF for sources can be easily extended to those of the obstacles, by simply fixing the minimum IIV for all the obstacle nodes and running the same Algorithm 1 or 2 at all other nodes. For example, in a 300×300 network, there are five source nodes and three obstacle nodes. We set the IIV of all sources as maximum 100, and fix all obstacles at minimum -50. The IPF for the five sources is shown in Fig. 5a, for three obstacles is shown in Fig. 5b, and the composite IPF is shown in Fig. 5c, which is a qualified information gradient. The no local maximum feature of the composite information gradient guarantees that all the sources can be reached wherever the sources are, and the mobile sinks can reach all the sources by following the ascending path of this information gradient.

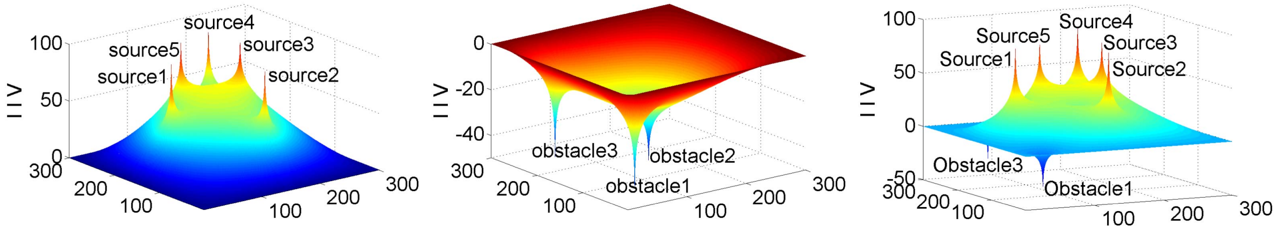
C. Energy-Balanced Navigation

Each node carries an IIV as the diffusion strength of EoIs from source node to the others, so as to form the smooth surface of the IPF. Then the hight difference of nodes provide the mobile sink with the gradient of the routing path. A very nice property of the IPF is to support a greedy routing strategy. Based on the IPF, the mobile sink at node u selects one of its neighbors $N(u)$, which has the largest IIV, to move up a step. In this way, every step of the route planning of the mobile sink is based only on the IIVs of the current node's neighbors, so that only local message exchange is sufficient. Without centralized data or routing decision, the route planning is completely in a distributed manner.

However, the energy balance problem of the entire network is still embedded in the gradient-based routing phase. As we all know, sensor nodes on the routing path of the mobile sink consume more energy for reception and transmission of the data. The lifetime of the network is directly influenced by the death of a sensor node [22]. Thus, we take the current residual energy of each node into consideration to plan the routes of mobile sinks. That is to say, we should make the gradient-based greedy routing path bypass the node which has low residual energy (less than a pre-defined threshold θ). To do this, during the construction and maintenance process of the IPF, the node which has low residual energy brings the hollow to the smooth surface of the IPF so that we pay attention to the distribution of the nodes' energy consumption. The attraction force of ascent on the gradient surface is caused by the next-hop node with higher residual energy while the repulsive force of the hollow is caused by the sensor with lower residual energy. Hence, the mobile sink is forced to climb the ascent of the gradient surface while bypassing the hollow with low residual energy. In addition, the movement steps of the mobile sink conduct towards a direction which is allocated according to the sensors gradient magnitude.

V. Evaluation

In this section, we discuss how to select an appropriate granularity of the skeleton node through simulation and evaluate the performance of the proposed algorithms.



(a) (b) (c)

Fig. 5. The composite IPF combines the IPF of five sources with the IPF of three obstacles. (a) IPF of sources. (b) IPF of obstacles. (c) Composite IPF.

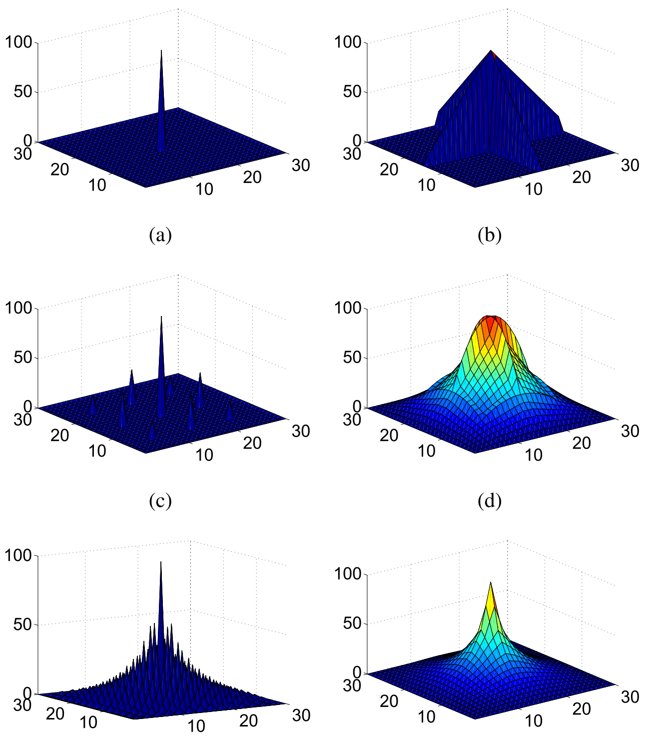


Fig. 6. The IPF under different granularities. Left {(a)(c)(e)}: The information potention field after performing the Jacobi iteration in the skeleton network. Right {(b)(d)(f)}: The IPF after interpolation process.

A. Proper Skeleton Node Selection

As mentioned before, in the HSCA it is important to select a set of skeleton nodes with proper granularity. We simulate several skeleton node selection schemes in a grid network of 20×20, with a source at the center of the network.

We compare the iterations needed for the whole network to get the ideal IPF under the following three schemes with different skeleton granularities,

- Scheme 1: A two-tier skeleton node structure. Select the source node and the perimeter nodes which are located in the same row or column as the source node to be the skeleton nodes.
- Scheme 2: A three-tier skeleton node structure. We set the source node as the innermost skeleton node and all perimeter nodes as the outmost layer skeleton nodes. Middle layer nodes are the ones with the same distance to the innermost layer and the outmost layer.
- Scheme 3: The hierarchy is determined by the scale of the network. Counting from the source node, we set a node as the skeleton node every two hops apart in line. The skeleton nodes are evenly distributed in the whole network.

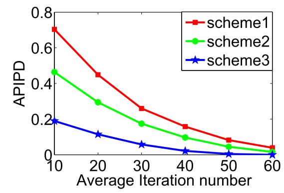


Fig. 7. The APIPD value of three skeleton schemes under different iteration numbers.

The IIV of source nodes are 100 and that of perimeter nodes are 0. The HSCA is used to build the IPF under different skeleton granularities. Fig. 6 shows the IPF evolution before the smoothing process step. We find that Scheme 3 provides a better IPF than the other two schemes, as shown in Fig. 6f.

To all of the above cases, we can finally obtain the IPF by the smoothing process in Algorithm 1. In Fig. 7, the x-axis represents the average number of iterations and the y-axis shows APIPD. The value of APIPD, which indicates the degree of approximation of the IPF to the harmonic function, reflects the precision of the IPF. It is shown that the precision of the IPF can be improved by more iterations. We also find that Scheme 3 leads to a significant reduction in the number of iterations, which results in a low energy consumption. Thus we suggest using Scheme 3 in HSCA to efficiently get the ideal IPF.

B. Energy-Efficient Construction of IPF

The energy consumption is a critical metric of the WSNs. As mentioned above, the energy consumption of communication leads to the main energy consumption during the IPF construction phase, and most communication load is caused by iterations. To demonstrate that both algorithms 1 and 2 can significantly decrease the energy consumption by reducing the number of iterations. This section compare the HSCA and VESA with the traditional Jacobi iteration method in terms of the number of iterations.

For different networks ranging from 60×60 to 120×120, we generate 2000 sources randomly for each network.

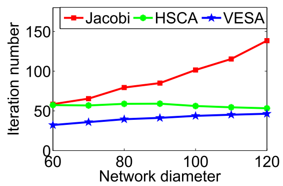


Fig. 8. The average number of iterations of three methods under different network diameter.

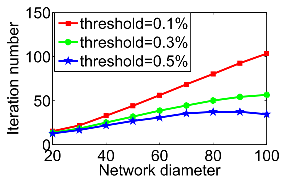


Fig. 9. The average number of iterations under different update threshold ε by the HSCA.

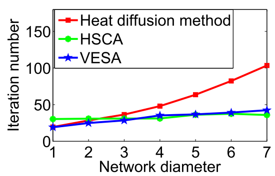


Fig. 10. The average number of iterations of three methods under different network density level.

We compare the performance of HSCA, VESA with the Jacobi method [7] in the context of the average number of iterations of all nodes. As shown in Fig. 10, the number of iteration of the heat diffusion method in [23] increases with the nodes density, but that of the proposed HSCA and VESA algorithm fluctuate slightly with the network density. The relative difference threshold is $\varepsilon = 0.1\%$, we compare the average number of iterations to establish the ideal IPF by using the HSCA, the VESA and the Jacobi iteration method. In Fig. 8, the x-axis denotes different network diameters and y-axis represents average number of iterations, which indicates the energy consumption. It is observed that the HSCA and the VESA can reduce the number of iterations by 80% compared with the traditional Jacobi iteration method for the same size network. We also find that the number of iterations increases with the network size. To show the influence of the update threshold ε on the average number of iterations, we compare the performance of three different thresholds using the HSCA. Fig. 9 shows that the average number of iterations decreases with the increase of the value of the updating threshold ε.

C. Evaluation of Prioritized IPF

Based on the prioritized IPF of five EoIs, the mobile sink is navigated to sources and achieve the global priority schedule. The mobile sinks follow the ascending path of the information gradient to event sources, which is guaranteed by selecting the neighbor node with the maximal IIV between all neighbors and route to it.

To validate the effectiveness of our design, we perform the following experiment. In this experiment, the size of the network increases from 20×20 to 100×100. We generate 1000 random events sequentially, and their appearance time

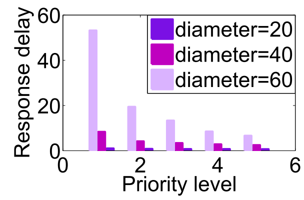


Fig. 11. Average response delay based on the prioritized IPF.

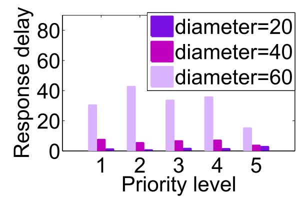


Fig. 12. Average response delay based on the ordinary IPF.

obeys the Poisson process model [24]. Each event belongs to a priority level which is classified to five levels $\{1, 2, 3, 4, 5\}$. Fig. 11 shows the average response delay between EoI's appearance and disappearance. It is observed that EoIs with higher priority level have a lower response delay time. The advantages of the prioritized IPF are more obvious in bigger networks, because the ratio of the response delay between high-priority EoIs and low-priority EoIs is even less. However, the average response delay is random based on the ordinary IPF without the priority consideration, as shown in Fig. 12.

D. Evaluation of Composite IPF

In order to demonstrate that the routes of the mobile sink can eventually reach sources and avoid obstacles only by making local decisions based on the composite IPF, we consider a 100×100 grid network with two sources and two obstacles in it. The position of two sources are $(10, 10)$ and $(20, 20)$, with an initial IIV of 100. The two obstacles are in $(23, 12)$ and $(16, 24)$. Assuming that the obstacles in this experiment are small, we set the initial IIV of two obstacles as a light strength -10, although it can be set as a much smaller value such as -50 indicating a larger obstacle.

The mobile sink is in the left corner, $(30, 2)$, of the network, and it moves each step by making a greedy decision. In detail, a mobile sink at or near the position of the sensor node u selects the direction of the one neighbor node which has the maximal IIV among all of its neighbors, expressed as $max\{\Phi(v), v \in N(u)\}$, to make the next move. The mobile sink is first guided to collect the source data in $(10, 10)$ by ascending the smooth surface of the composite IPF, as shown in Fig. 13. Then the diffused information of the source in $(10, 10)$ is deleted once it is collected by the mobile sink. As shown in Fig. 13, red tringles show the route path of the mobile sink, which reaches sources and avoids obstacles by simply going to the neighbor with the most significant difference in IIV, based on the composite IPF of two sources and two obstacles. In state-of-the-art distributed routing protocols to avoid obstacles [25], the MDC may not move along straight lines. Compared with the distributed routing method in [26], its routing path exists a larger collision.

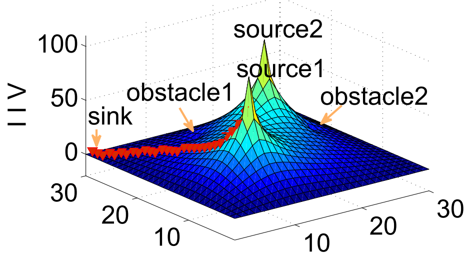


Fig. 13. The mobile sink routes to sources and avoids obstacle (23, 12).

VI. Related Work

In a highly dynamic network, the cost and time required to fetch information increases as the Euclidean distance from the object increases. As shown in [27], the gradient-based routing is a reserved local predictive standard where every node figures the height of the node. The concept of information potential helps in minimizing this cost. The work [28] proposed an architecture for content search in a mobile opportunistic network based on the information gradient, and query forward with the aim to minimize the search time and transmission overhead. Liu *et al.* [29] presented an Energy-Balancing unequal Clustering Approach for Gradient-based routing (EBCAG) in wireless sensor networks to improve the network lifetime. Kannan and Paramasivan [30] propose a gradient-based network for achieving the optimal path which reduces energy consumption of sensor nodes. Wei and Qi [23] used heat diffusion equation to aid information querying and navigation in a dynamic and real-time environment. They also presented a multi-scale gradient descent method to satisfy users requirements in WSNs. The work [31] proposed a gravity gradient routing concept that prioritizes the areas of high user-presence to combine the targeted delivery and resilience to potential sensor-load heterogeneity within the network. The path following process based on the six degrees was introduced to verify the practicability of the method of artificial potential field (APF) UAV path planning method in [32]. The work [33] presented a method based on a diffusion equation to accomplish the navigation process conveniently and efficiently in complex environments.

Reducing the energy consumption and prolonging the network lifetime are critical issues to construct the IPF. Many efforts [20]–[22] have been made to achieve these goals. Gao *et al.* [34] allowed a smooth IPF and enabled local search algorithms to work well. A sampled scalar field can be stored in this distributed fashion, with only a modest amount of additional storage and network traffic. Sarkar *et al.* [35] proposed to construct multi-resolution data representations for WSNs. The algorithm is low-cost in communication and roubust to link failures. Huang *et al.* [36] presented a state-free gradient-based forwarding (SGF) protocol to address challenges of resource constraints.

Recently sink mobility has been exploited in numerous schemes to prolong the lifetime of WSNs. Paper [37] identified several advantages of sink mobility: sensor lifetime enhancement, improved coverage, improved throughput. Kinalis *et al.* [2] proposed a biased, adaptive sink mobility scheme to improve energy-latency trade-offs. Madhumathy and Sivakumar [38] designed a technique by which the sensor node encodes the resulting block and communicates to the sink before the data is ready for transmission. Paper [39] focused on the impact of selecting mobile sink and the sojourn positions, and found that both selections were of importance.

VII. Conclusion

With the information potential field (IPF), the query navigation or routing of data packets can achieve global goals based on local decision. In a word, leveraging the Jacobi iteration method to construct the IPF has suffered from high energy consumption, low convergence speed and non-convergence at certain sensors in a large-scale network. This paper is an attempt to revisit the basic Jacobi iteration method, and demonstrate two energy-efficient IPF construction algorithms: HSCA and VESA. Both algorithms reduce the number of iterations, save energy consumption and have good performance in large-scale WSNs. Towards a more practical solution, we take more actual requirement into account to construct two kinds of complex IPF, i.e., prioritized IPF and composite IPF of sources and that of obstacles. Comprehensive simulation results show the feasibility of the proposed algorithms, which can reduce iterations to reach a convergence status by 80% and conserve 30-50% energy consumption on average.

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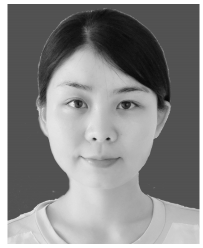

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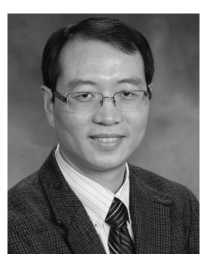

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